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then permutation of 2, 3, 5, 6

$${}^4P_4 = \frac{4!}{(4-4)!}$$

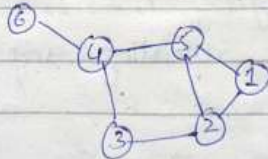
$$\frac{4 \cdot 3 \cdot 2 \cdot 1}{0!} \rightarrow \frac{4 \cdot 3 \cdot 2 \cdot 1}{1}$$

24.

21 is fixed at extreme left then permutation of 3, 5, 6.

(25) Element of graph theory.

This article is about sets of vertices connected by edges.



A graph with six vertices & seven edges.

These edges are directed and undirected.

→ If the vertices represent people at a party, and

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there is an edge b/w two people if they shake hands then this graph is undirected because any person A can shake hand with a person B only if B also shake hand with person A.

In contrast if person A owes money from person B then it is directed

This former type of graph is called an undirected graph while the latter type of graph is directed graph.

Definition:

A graph (denoted as $G = (V, E)$) consist of a non-empty set of vertices or nodes V and a set of edge E .

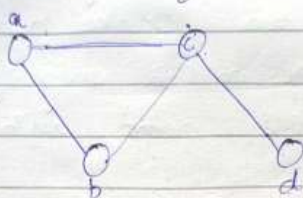
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e.g:

$$G = (V, E)$$

$$V = \{a, b, c, d\}$$

$$E = \{(a, b), (a, c), (b, c), (c, d)\}$$

**Degree of vertex**

The degree of a vertex V of a graph G (denoted by $\deg V$) is the number of edges incident with the vertex V .

Vertex	Degree	Even/Odd.
a	2	Even
b	2	Even
c	3	Odd
d	1	Odd

Even & odd vertex:

If the degree of a vertex is even, then vertex is called as even vertex

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and if degree is odd then called odd vertex.

Degree of graph:

The degree of a graph is largest vertex of that graph.

Handshaking Lemma:

In graph the sum of all degree of all the vertices is equal to the twice the number of edges. [Theorem - $\sum \deg(V) = 2e$]

TYPES OF GRAPH

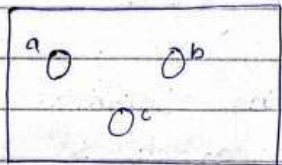
These are different types of graph which we discuss

- Null graph
- Simple
- Multi
- Directed & undirected
- Connected & Disconnected
- Regular
- Complete
- Cycle.

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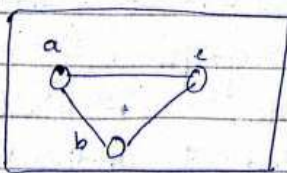
• Null graph:

A null graph have no edges. The null graph of n vertices is denoted by N_n .



• Simple graph:

A graph is called simple graph / strict graph if the graph is undirected and does not contain any loop or multi edges.

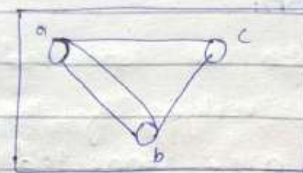


• Multi graph:

If in a graph multiple

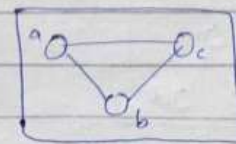
143-

edges b/w the same set of vertices are allowed. it is called multigraph. In other words, it is a graph having at least one loop or multiple edges.

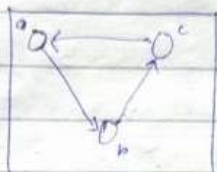


• Directed & undirected graph:

A graph $G = (V, E)$ is called a directed graph if the edges set is made of ordered vertex pair and a graph is called undirected if the edges set is made of unordered vertex pair.



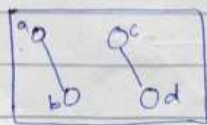
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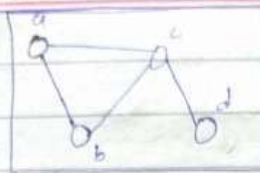
• Connected & disconnected graph:

A graph is connected if any two vertices of the graph are connected by a path; while a graph is disconnected if at least two vertices of the graph are not connected by a path.

If a graph G is disconnected then every maximal connected subgraph of G is called a connected component of graph G .

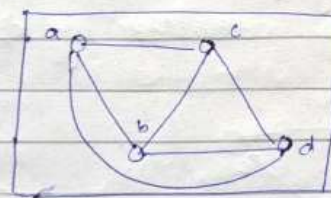


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• Regular graph:

A regular graph is if all the vertices of the graph have same degree. In a regular graph G of degree ' r ', the degree of each vertex of ' G ' is ' r '.

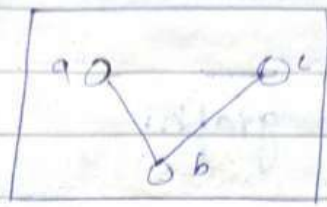


• Complete graph:

A graph is called complete graph if every two vertices pair are joined by exactly one edges. The complete graph

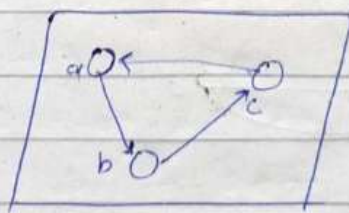
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with 'n' vertices is denoted by K_n .



• Cycle graph:

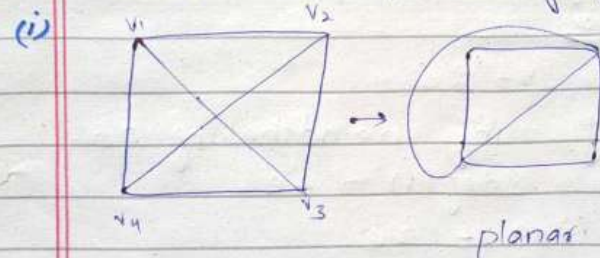
If a graph consists of a single cycle it is called cycle graph. The cycle graph with n vertices is denoted by C_n .



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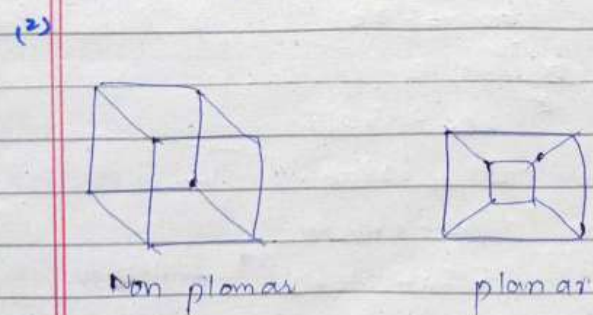
(26) Planar graph:

A graph is called planar graph if it can be drawn in plane without any edge crossing and a graph that cannot be drawn on a plane without a cross over b/w its edges is called non-planar graph.



Non planar

planar



Non planar

planar

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(28) Euler formula:

$$r = e - v + 2$$

Handshaking theorem:

$$\sum \deg(v) = 2e$$

Q:1

If there are 20 vertices each of degree 3, then into how many region does of this planar graph splits the plane.

Sol

$$v = 20$$

By using handshaking for-

$$\sum \deg(v) = 2e$$

$$3(20) = 2e$$

$$20 + 20 + 20 = 2e$$

$$60 = 2e$$

$$60^{30} = e$$

$$e = 30$$

So put value.

$$r = e - v + 2$$

$$r = 30 - 20 + 2$$

$$r = 12$$

20 is plus three time

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edge formula:

$$e = \frac{n(n-1)}{2}$$

Result - 1:

If G is a connected planar simple graph with 'e' edge & 'v' vertices. where $v \geq 3$ then $e \leq 3v - 6$

Is K_5 planar?

$$\frac{n(n-1)}{2} = e$$

$$\frac{5(5-1)}{2} = e$$

$$\frac{5(4)}{2} = e$$

$$\frac{20}{2} = e$$

$$e = 10$$

So,

$$v = 5$$

$$e = 10$$

using result.

$$e \leq 3v - 6$$

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$$10 \leq 3v - 6$$

$$10 \leq 3(5) - 6$$

$$10 \leq 15 - 6$$

$$10 \leq 9$$

It is not planar.

Q.2

$$\frac{n(n-1)}{2} = \text{edg.}$$

$$\frac{7(7-1)}{2} = \text{edg.}$$

$$\frac{2(6)}{2} = \text{edg.}$$

$$\frac{4 \times 21}{2} = \text{edg.}$$

$$= 21$$

So,

$$v = 7$$

$$e = 21.$$

formula,

$$e \leq 3v - 6$$

$$21 \leq 3(7) - 6$$

$$21 \leq 21 - 6$$

$$21 \leq 15$$

It is not planar.

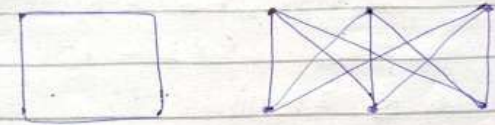
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Result. 2

If a connected planar simple graph has e edge and v vertices with $v \geq 3$ and no circuits of length 3 then $e \leq 2v - 4$

Question.

① K_3 , 3 planar?



$$v = 3 + 3 = 6$$

$$e = 3 \times 3 = 9.$$

As result.

$$e \leq 2v - 4$$

$$9 \leq 2(6) - 4$$

$$9 \leq 12 - 4$$

$$9 \leq 8$$

It is not planar.

② K_2 , 16 = ?

$$v = 2 + 6$$

$$v = 8$$

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$$e = 6 \times 2$$

$$= 12$$

As result-

$$e \leq 2v - 4$$

$$12 \leq 2(8) - 4$$

$$12 \leq 16 - 4$$

$$12 \leq 12$$

It is planar.

③ K_3 , 5 = ?

$$v = 5 + 3 = 8$$

$$e = 5 \times 3 = 15$$

$$e \leq 2v - 4$$

$$15 \leq 2(8) - 4$$

$$15 \leq 16 - 4$$

$$15 \leq 12$$

It is not planar.

④ K_{12} , 6 = ?

$$v = 12 + 6 = 18$$

$$e = 12 \times 6 = 72$$

As result

$$e \leq 2v - 4$$

$$72 \leq 2(18) - 4$$

$$72 \leq 36 - 4$$

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$$72 \leq 32$$

It is not planar.

⑤ K_{13} , 2 = ?

$$v = 13 + 2 = 15$$

$$e = 13 \times 2 = 26$$

$$e \leq 2v - 4$$

$$26 \leq 2(15) - 4$$

$$26 \leq 30 - 4$$

$$26 \leq 26$$

It is planar.

⑥ K_5 , 6 = ?

$$v = 5 + 6 = 11$$

$$e = 5 \times 6 = 30$$

$$e \leq 2v - 4$$

$$30 \leq 2(11) - 4$$

$$30 \leq 22 - 4$$

$$30 \leq 18$$

It is not planar.

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Q9) Hamiltonian path.

Path:-

Hamiltonian path contains each vertex exactly once

Hamiltonian circuit:

First & last vertex are same.

e.g:

[In a row no. doesn't repeat]



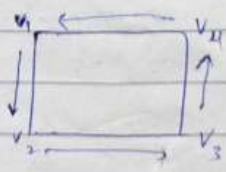
$v_1 \rightarrow v_2 \rightarrow v_3 \rightarrow v_4 \rightarrow v_5$

length of ham-path = $n-1$

$= 5-1$

$= 4$ (edge).

[No. of row is repeat once if last are equal]



$v_1 \rightarrow v_2 \rightarrow v_3 \rightarrow v_4$

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length = edges

$l_1 = 4$

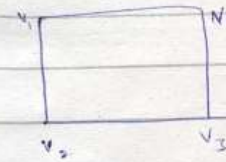
Q.1 Theorem:

(H. path)

Let G_1 be a graph of 'n' vertices then 'G' has ham-path if for only two vertices 'u' & 'v' of G_1

Sol

$\text{Deg}(u) + \text{Deg}(v) \geq n$



$n=4$

Let v_3 & v_4

$\text{Deg } v_3 = 2$

$\text{Deg } v_4 = 2$

So,

$2+2 \geq 4$

$4 \geq 4$

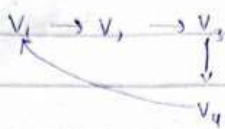
Q.2

(H. circuit)

On removing any one edge from a hamiltonian circuit, then we are left

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with hamiltonian path.



length = edge

$4 = 4$

$\text{Deg}(u) + \text{Deg}(v) \geq n$

$\text{Deg } v_3 = 2$

$\text{Deg } v_4 = 2$

So,

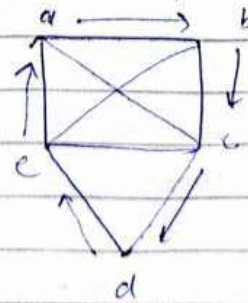
$2+2 \geq 4$

$4 \geq 4$

Q.3

which is ham-path & ham-circuit

(i)

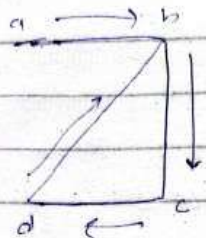


$a \rightarrow b \rightarrow c \rightarrow d \rightarrow e \rightarrow a$

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It is ham-circuit.

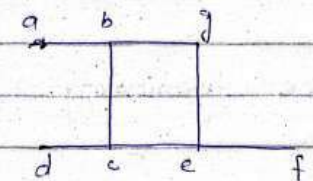
(ii)



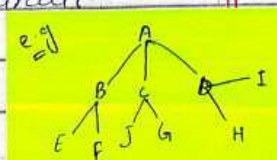
It is ham-path.

$a \rightarrow b \rightarrow c \rightarrow d \rightarrow b$

(iii)



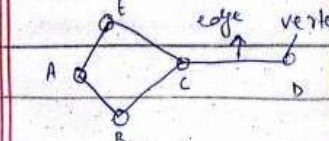
It is ham-circuit.



(30)

Tree

A tree is a graph i.e. a graph having its edges are branches.



No. of node = 10

No. of edge = 9.

Parent $\rightarrow$ A

child $\rightarrow$ B, C, D

Sibling $\rightarrow$ Acc to A

B, C, D

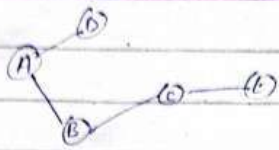
Leaf node (external)

E, F, J, G, H, I

Internal node (Internal)

A, B, C, D

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**Rooted tree:-**

A rooted tree is a tree in which a particular vertex is distinguished from the others and is called the root.

Tree terminology.**Root:**

The first node in a tree is called often the top level node.

Edge

The connecting link between any two nodes is called as edge. In a tree with 'N' number of nodes there will be a maximum of $N-1$ number of edges.

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Parent:

The node which is predecessor of any node is called as parent node.

Child:

The node which is descendant of any node is called as child node.

Sibling:

Nodes which belong to same parent are called as siblings.

Leaf node:

The node which does not have a child as node with degree zero is called as leaf node. Leaf node is called external / terminal node.

Internal node,

The node which has at least one child is called as internal node.

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Degree:

The total number of children of a node or number of subtree of a node is called as degree of node.

Level:

The root node is said to be at level 0 and the children of root node are at level 1 and the children of the nodes which are at level 1 will be at level 2 and so on.

Path:

The sequence of nodes and edges from one node to another node is called as path b/w that two nodes. Length of a path is total number of nodes in that path.

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Subtree:

Each child from a node forms a subtree recursively. Every child node will form a subtree on its parent node.

(31) Tree traversal:

A traversal of a tree is a process to traverse a tree in a systematic way so that each vertex is visited exactly once.

=> (i) Preorder traversal:

The preorder traversal of a binary tree is defined recursively as.

1. Visit the root.
2. Traverse the left subtree in preorder.
3. Traverse the right subtree in preorder.

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⇒ (2) Postorder traversal:

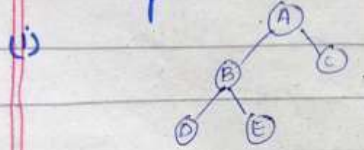
The postorder traversal of a binary tree is defined recursively as.

1. Traverse the left subtree in postorder.
2. Traverse the right subtree in postorder.
3. Visit the root.

⇒ (3) Inorder traversal:

The inorder traversal of a binary tree is defined recursively as.

1. Traverse in inorder the left subtree.
2. Visit the root.
3. Traverse in inorder the right subtree.

Examples:

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Preorder (L.S)

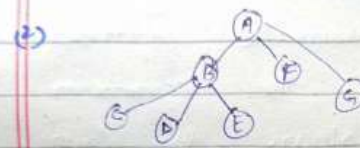
A, B, D, E

Postorder (R.S)

A, E, B, C, A

In-order (Center)

D, B, E, A, C

**Preorder (L.S)**

A, B, C, D, E, F, G

Post order (R.S)

C, D, E, B, F, G, A

In-order (Center)

C, D, E, B, F, G, A

(27) Graph colouring

The assign of colours to the vertices of G one colour to each vertex, so the adjacent vertices are assigned

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different colour is the proper colouring of G (graph) or simply vertex coloring. If G has 'n' colouring, then G is said to be 'n' colourable.

Chromatic number:

The chromatic number of graph G is the minimum number of colours to colour the vertices of the graph G and is denoted by $\chi(G)$.



$$\chi(G) = 3$$



$$\chi(G) = 4$$

Q: Every Bipartite graph is 2 coloured.

Let $\{V_1, V_2, V_3, V_4, V_5\}$



$$\chi(G) = 2$$

that does not have edge on same line.

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(21) Counting:-

- How many different 8-letter passwords are there?
- How many possible way are there to pick 11 soccer players out of a 20-players team?
- Most importantly, counting is a basic for computing probabilities of discrete event.

Basic counting principle:

Sum rule If we have tasks $T_1, T_2, T_3, \dots, T_m$ that can be done in $n_1, n_2, \dots, n_m$ ways, respectively, and no two of these tasks can be done at the same time, then there are $n_1 + n_2 + \dots + n_m$ way to do one of these tasks.

The product rule:-